

Phenolic content of virgin olive oil improves ischemic reactive hyperemia in hypercholesterolemic patients.



OBJECTIVES: The goal of this study was to evaluate the effects of the phenolic content of virgin olive oil on endothelial reactivity. **BACKGROUND:** Endothelial-dependent vasodilatation is impaired during the postprandial state, and oxidative stress could play a key role in its development. **METHODS:** Twenty-one hypercholesterolemic volunteers received two breakfasts, using a randomized sequential crossover design. Both arms received the same olive oil, but one had its phenolic acid content reduced from 400 to 80 ppm. Ischemic reactive hyperemia (IRH) was measured with a laser-Doppler procedure at baseline and 2 h and 4 h after oil intake. Postprandial plasma concentrations of lipid fractions, lipoperoxides (LPO), 8-epi prostaglandin-F₂alpha, and nitrates/nitrites (NO(x)) were obtained at baseline and after 2 h of the fat meal. **RESULTS:** The intake of the polyphenol-rich breakfast was associated with an improvement in endothelial function, as well as a greater increase in concentrations of NO(x) ($p < 0.001$) and a lower increase in LPO ($p < 0.005$) and 8-epi prostaglandin-F₂alpha ($p < 0.001$) than the ones induced by the low polyphenol fat meal. A positive correlation was found to exist between NO(x) and enhanced endothelial function at the second hour ($r = 0.669$; $p < 0.01$). Furthermore, a negative correlation was found between IRH and LPO ($r = -0.203$; $p < 0.05$) and 8-epi prostaglandin-F₂alpha levels ($r = -0.440$; $p < 0.05$). **CONCLUSIONS:** A meal containing high-phenolic virgin olive oil improves ischemic reactive hyperemia during the postprandial state. This phenomenon might be mediated via reduction in oxidative stress and the increase of nitric oxide metabolites.